

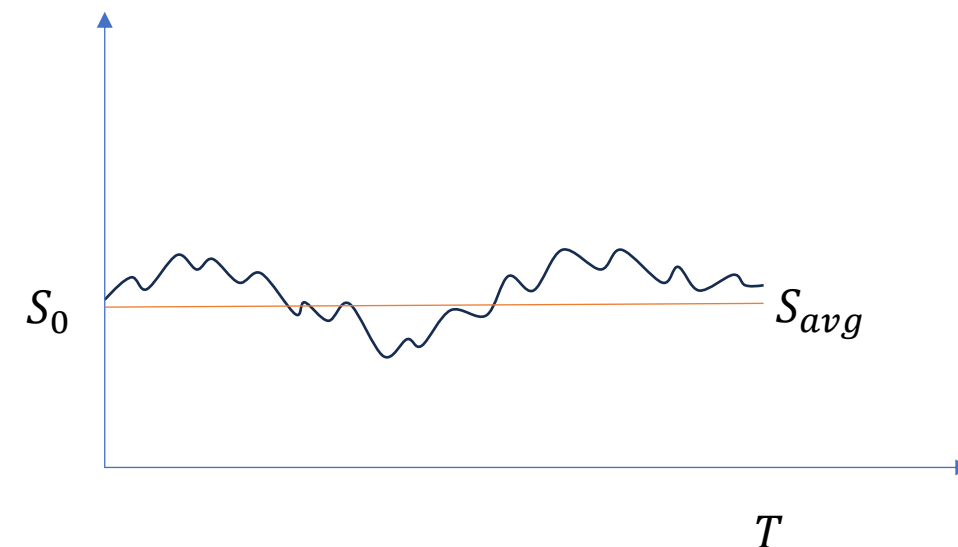
Asian Options

(1. Geometric Averaging)

Asian Options

An **Asian option** is one whose payoff includes a time average of the underlying asset price. The Asian options can vary depending on the key dimensions below

- Sampling method (Continuous or Discrete)
- Averaging (Arithmetic or Geometric)
- Strike (Fixed or Floating)



Fixed Strike

Below are the specifications of different sampling and averaging schemes under Fixed Strike Call Option

Discrete Sampling

$$\max \left\{ \frac{1}{n} \sum_{i=1}^n S_{t_i} - K, 0 \right\} \quad (\text{Arithmetic})$$

$$\max \left\{ \prod_{i=1}^n (S_{t_i})^{\frac{1}{n}} - K, 0 \right\} \quad (\text{Geometric})$$

Continuous Sampling

$$\max \left\{ \frac{1}{T} \int_0^T S_u du - K, 0 \right\} \quad (\text{Arithmetic})$$

$$\max \left\{ \exp \left(\frac{1}{T} \int_0^T \ln(S_u) du \right) - K, 0 \right\} \quad (\text{Geometric})$$

Floating Strike

For Floating Strike Call Option

Discrete Sampling

$$\max \left\{ S_T - \frac{1}{n} \sum_{i=1}^n S_{t_i}, 0 \right\} \quad (\text{Arithmetic})$$

$$\max \left\{ S_T - \prod_{i=1}^n (S_{t_i})^{\frac{1}{n}}, 0 \right\} \quad (\text{Geometric})$$

Continuous Sampling

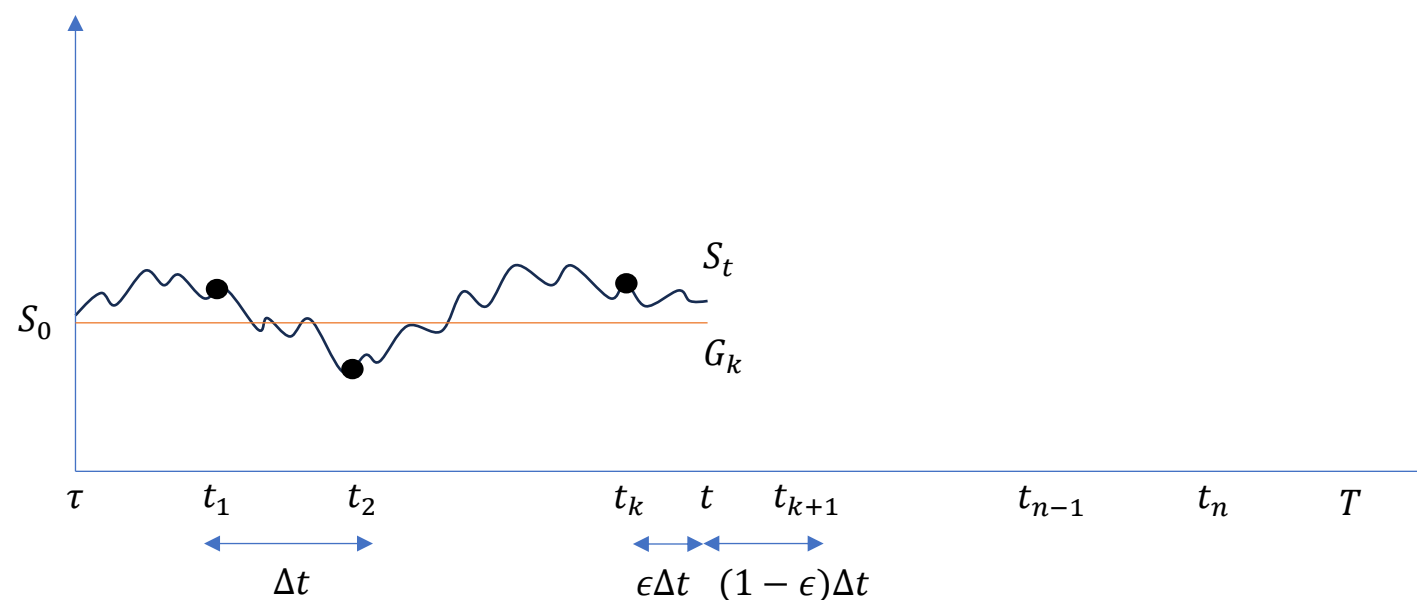
$$\max \left\{ S_T - \frac{1}{T} \int_0^T S_u du, 0 \right\} \quad (\text{Arithmetic})$$

$$\max \left\{ S_T - \exp \left(\frac{1}{T} \int_0^T \ln(S_u) du \right), 0 \right\} \quad (\text{Geometric})$$

Pricing the discrete geometric(I)

Asian options typically do not have closed form solutions except the one that uses Geometric average. Let's derive the analytical pricing formula for the same.

Consider the sampling window $[\tau, T]$ with sampling dates $\tau \leq t_1 < t_2 < \dots < t_n \leq T$ and $S_{t_1}, S_{t_2}, \dots, S_{t_n}$ are the asset prices on the sampling dates. Suppose, we are currently at t , and the stock price is S_t . We have observed $S_{t_1}, S_{t_2}, \dots, S_{t_k}$.



Pricing the discrete geometric(II)

The final underlier is given as

$$\begin{aligned}
 G_n(\tau, T) &= \left(\prod_{i=1}^n S_{t_i} \right)^{\frac{1}{n}} = \left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} \left(\prod_{i=k+1}^n S_{t_i} \right)^{\frac{1}{n}} = \left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} (S_{t_n} S_{t_{n-1}} S_{t_{n-2}} \dots S_{t_{k+2}} S_{t_{k+1}})^{\frac{1}{n}} \\
 &= \underbrace{\left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}}}_{\text{(known)}} S_t^{\frac{n-k}{n}} \left[\left(\frac{S_{t_n}}{S_{t_{n-1}}} \right) \left(\frac{S_{t_{n-1}}}{S_{t_{n-2}}} \right)^2 \left(\frac{S_{t_{n-2}}}{S_{t_{n-3}}} \right)^3 \dots \left(\frac{S_{t_{k+1}}}{S_t} \right)^{n-k} \right]^{\frac{1}{n}}
 \end{aligned}$$

For $i = k + 2, k + 3, \dots, n - 1, n$ under risk-neutral measure $\mathbb{Q}$,

$$\log \left(\frac{S_{t_i}}{S_{t_{i-1}}} \right) = \left(r - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma (W_{t_i} - W_{t_{i-1}}) \quad \text{and}$$

$$\log \left(\frac{S_{t_{k+1}}}{S_t} \right) = \left(r - \frac{1}{2} \sigma^2 \right) (1 - \epsilon) \Delta t + \sigma (W_{t_{k+1}} - W_t)$$

Pricing the discrete geometric(III)

Taking log

$$\log(G_n(\tau, T)) = \log \left[\left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} S_t^{\frac{n-k}{n}} \right] + \frac{1}{n} \left[\log \left(\frac{S_{t_n}}{S_{t_{n-1}}} \right) + 2 \log \left(\frac{S_{t_{n-1}}}{S_{t_{n-2}}} \right) + \dots + (n-k) \log \left(\frac{S_{t_{k+1}}}{S_t} \right) \right]$$

$$m = \mathbb{E}^{\mathbb{Q}}[\log(G_n(\tau, T))] = \log \left[\left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} S_t^{\frac{n-k}{n}} \right] + \frac{1}{n} \left(r - \frac{1}{2} \sigma^2 \right) \Delta t (1 + 2 + 3 + \dots (n-k-1) + (n-k)(1-\epsilon))$$

$$= \log \left[\left(\prod_{i=1}^k S_{t_i} \right)^{\frac{1}{n}} S_t^{\frac{n-k}{n}} \right] + \left(r - \frac{1}{2} \sigma^2 \right) \Delta t \left[\frac{(n-k-1)(n-k)}{2n} + \frac{(n-k)(1-\epsilon)}{n} \right]$$

$$s^2 = \mathbb{V}^{\mathbb{Q}}[\log(G_n(\tau, T))] = \frac{\sigma^2 \Delta t}{n^2} (1^2 + 2^2 + \dots (n-k-1)^2 + (n-k)^2(1-\epsilon))$$

$$= \sigma^2 \Delta t \left(\frac{(n-k-1)(n-k)(2n-2k-1)}{6n^2} + \frac{(n-k)^2(1-\epsilon)}{n^2} \right)$$

Pricing the discrete geometric(IV)

Log of geometric average follows normal distribution,

$$\log(G_n(\tau, T)) \sim N(m, s^2)$$

So, the geometric average follows lognormal distribution just like the stock price itself. Because of this property, we can write down the call option price in BSM style

$$\begin{aligned} C_{\text{fixed strike}}^{(g)}(S_t, G_k, t; K, T) &= e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[\max\{G_n(\tau, T) - K, 0\} | \mathcal{F}_t] \\ &= e^{-r(T-t)} \left[e^{m + \frac{1}{2}s^2} \Phi(d_+) - K \Phi(d_-) \right] \end{aligned}$$

$$d_+ = \frac{m - \ln K + s^2}{s} \quad d_- = d_+ - s$$